Random Forest

Libraries

```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.
0.0 ──
## ✓ dplyr     1.2.1     ✓ readr     2.2.0
## ✓ forcats   1.0.1     ✓ stringr   1.6.0
## ✓ ggplot2   4.0.3     ✓ tibble    3.3.1
## ✓ lubridate 1.9.5     ✓ tidyr     1.3.2
## ✓ purrr     1.2.2
## ── Conflicts ────────────────────────────────────────── tidyverse_conflict
s() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
 conflicts to become errors
```

```
library(tidymodels)
```

```
## ── Attaching packages ─────────────────────────────────── tidymodels 1.
5.0 ──
## ✓ broom        1.0.12     ✓ rsample      1.3.2
## ✓ dials        1.4.3      ✓ tailor       0.1.0
## ✓ infer        1.1.0      ✓ tune         2.1.0
## ✓ modeldata    1.5.1      ✓ workflows    1.3.0
## ✓ parsnip      1.6.0      ✓ workflowsets 1.1.1
## ✓ recipes      1.3.2      ✓ yardstick    1.4.0
## ── Conflicts ────────────────────────────────────── tidymodels_conflict
s() ──
## ✖ scales::discard() masks purrr::discard()
## ✖ dplyr::filter()   masks stats::filter()
## ✖ recipes::fixed()  masks stringr::fixed()
## ✖ dplyr::lag()      masks stats::lag()
## ✖ yardstick::spec() masks readr::spec()
## ✖ recipes::step()   masks stats::step()
```

```
library(mice)
```

```
##
## Attaching package: 'mice'
##
## The following object is masked from 'package:stats':
##
##     filter
##
```

```
## The following objects are masked from 'package:base':
##
##     cbind, rbind
```

```
library(VIM)
```

```
## Loading required package: colorspace
## Loading required package: grid
## Registered S3 method overwritten by 'car':
##   method           from
##   na.action.merMod lme4
## VIM is ready to use.
##
## Suggestions and bug-reports can be submitted at: https://github.com/statis
tikat/VIM/issues
##
## Attaching package: 'VIM'
##
## The following object is masked from 'package:recipes':
##
##     prepare
##
## The following object is masked from 'package:datasets':
##
##     sleep
```

```
library(ranger)
library(randomForest)
```

```
## randomForest 4.7-1.2
## Type rfNews() to see new features/changes/bug fixes.
##
## Attaching package: 'randomForest'
##
## The following object is masked from 'package:ranger':
##
##     importance
##
## The following object is masked from 'package:dplyr':
##
##     combine
##
## The following object is masked from 'package:ggplot2':
##
##     margin
```

```
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
```

```
##
## The following objects are masked from 'package:yardstick':
##
##     precision, recall, sensitivity, specificity
##
## The following object is masked from 'package:rsample':
##
##     calibration
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(skimr)
library(GGally)
library(gridExtra)
```

```
##
## Attaching package: 'gridExtra'
##
## The following object is masked from 'package:randomForest':
##
##     combine
##
## The following object is masked from 'package:dplyr':
##
##     combine
```

```
library(vip)
```

```
##
## Attaching package: 'vip'
##
## The following object is masked from 'package:utils':
##
##     vi
```

Load dataset

```
train <- read_csv("C:/Users/miche/OneDrive/Desktop/BAN 502 Predictive Analytics/Course Project/train.csv")
```

```
## Rows: 26570 Columns: 26
## ── Column specification ────────────────────────────────────────────────────
## Delimiter: ","
## chr  (4): product_code, attribute_0, attribute_1, failure
## dbl (22): id, loading, attribute_2, attribute_3, measurement_0, measurement_...
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
```

```
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this m
essage.
```

Structure and summary

```
str(train)
```

```
## spc_tbl_ [26,570 × 26] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ id            : num [1:26570] 0 1 2 3 4 5 6 7 8 9 ...
##  $ product_code  : chr [1:26570] "A" "A" "A" "A" ...
##  $ loading       : num [1:26570] 80.1 84.9 82.4 101.1 188.1 ...
##  $ attribute_0   : chr [1:26570] "material_7" "material_7" "material_7" "m
aterial_7" ...
##  $ attribute_1   : chr [1:26570] "material_8" "material_8" "material_8" "m
aterial_8" ...
##  $ attribute_2   : num [1:26570] 9 9 9 9 9 9 9 9 9 9 ...
##  $ attribute_3   : num [1:26570] 5 5 5 5 5 5 5 5 5 5 ...
##  $ measurement_0 : num [1:26570] 7 14 12 13 9 11 12 4 9 10 ...
##  $ measurement_1 : num [1:26570] 8 3 1 2 2 4 2 8 6 4 ...
##  $ measurement_2 : num [1:26570] 4 3 5 6 8 0 4 8 5 7 ...
##  $ measurement_3 : num [1:26570] 18 18.2 18.1 17.3 19.3 ...
##  $ measurement_4 : num [1:26570] 12.5 11.5 11.7 11.2 12.9 ...
##  $ measurement_5 : num [1:26570] 15.7 17.7 16.7 18.6 17 ...
##  $ measurement_6 : num [1:26570] 19.3 17.9 18.2 18.3 15.7 ...
##  $ measurement_7 : num [1:26570] 11.7 12.7 12.7 12.6 11.3 ...
##  $ measurement_8 : num [1:26570] 20.2 17.9 18.3 19.1 18.1 ...
##  $ measurement_9 : num [1:26570] 10.7 12.4 12.7 12.5 10.3 ...
##  $ measurement_10: num [1:26570] 15.9 17.9 15.6 16.3 17.1 ...
##  $ measurement_11: num [1:26570] 17.6 17.9 NA 18.4 19.9 ...
##  $ measurement_12: num [1:26570] 15.2 11.8 13.8 10 12.4 ...
##  $ measurement_13: num [1:26570] 15 14.7 16.7 15.2 16.2 ...
##  $ measurement_14: num [1:26570] NA 15.4 18.6 15.6 12.8 ...
##  $ measurement_15: num [1:26570] 13 14.4 14.1 16.2 13.2 ...
##  $ measurement_16: num [1:26570] 14.7 15.6 17.9 17.2 16.4 ...
##  $ measurement_17: num [1:26570] 764 682 663 826 580 ...
##  $ failure       : chr [1:26570] "No" "No" "No" "No" ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   id = col_double(),
##   ..   product_code = col_character(),
##   ..   loading = col_double(),
##   ..   attribute_0 = col_character(),
##   ..   attribute_1 = col_character(),
##   ..   attribute_2 = col_double(),
##   ..   attribute_3 = col_double(),
##   ..   measurement_0 = col_double(),
##   ..   measurement_1 = col_double(),
##   ..   measurement_2 = col_double(),
##   ..   measurement_3 = col_double(),
##   ..   measurement_4 = col_double(),
```

```
##   ..   measurement_5 = col_double(),
##   ..   measurement_6 = col_double(),
##   ..   measurement_7 = col_double(),
##   ..   measurement_8 = col_double(),
##   ..   measurement_9 = col_double(),
##   ..   measurement_10 = col_double(),
##   ..   measurement_11 = col_double(),
##   ..   measurement_12 = col_double(),
##   ..   measurement_13 = col_double(),
##   ..   measurement_14 = col_double(),
##   ..   measurement_15 = col_double(),
##   ..   measurement_16 = col_double(),
##   ..   measurement_17 = col_double(),
##   ..   failure = col_character()
##   .. )
##  - attr(*, "problems")=<pointer: 0x000002205a4f56e0>
```

```
summary(train)
```

```
##        id            product_code      loading           attribute_0
##  Min.   :    0   Length   :26570   Min.   : 33.16   Length   :26570
##  1st Qu.: 6642   N.unique :    5   1st Qu.: 99.99   N.unique :    2
##  Median :13284   N.blank  :    0   Median :122.39   N.blank  :    0
##  Mean   :13284   Min.nchar:    1   Mean   :127.83   Min.nchar:   10
##  3rd Qu.:19927   Max.nchar:    1   3rd Qu.:149.15   Max.nchar:   10
##  Max.   :26569                     Max.   :385.86
##                                    NAs    :250
##     attribute_1     attribute_2     attribute_3   measurement_0
##  Length   :26570   Min.   :5.000   Min.   :5.00   Min.   : 0.000
##  N.unique :    3   1st Qu.:6.000   1st Qu.:6.00   1st Qu.: 4.000
##  N.blank  :    0   Median :6.000   Median :8.00   Median : 7.000
##  Min.nchar:   10   Mean   :6.754   Mean   :7.24   Mean   : 7.416
##  Max.nchar:   10   3rd Qu.:8.000   3rd Qu.:8.00   3rd Qu.:10.000
##                    Max.   :9.000   Max.   :9.00   Max.   :29.000
##
##  measurement_1    measurement_2    measurement_3   measurement_4
##  Min.   : 0.000   Min.   : 0.000   Min.   :13.97   Min.   : 8.008
##  1st Qu.: 5.000   1st Qu.: 4.000   1st Qu.:17.12   1st Qu.:11.051
##  Median : 8.000   Median : 6.000   Median :17.79   Median :11.733
##  Mean   : 8.233   Mean   : 6.257   Mean   :17.79   Mean   :11.732
##  3rd Qu.:11.000   3rd Qu.: 8.000   3rd Qu.:18.47   3rd Qu.:12.410
##  Max.   :29.000   Max.   :24.000   Max.   :21.50   Max.   :16.484
##                                    NAs    :381     NAs    :538
##  measurement_5   measurement_6   measurement_7    measurement_8
##  Min.   :12.07   Min.   :12.71   Min.   : 7.968   Min.   :15.22
##  1st Qu.:16.44   1st Qu.:16.84   1st Qu.:11.045   1st Qu.:18.34
##  Median :17.13   Median :17.52   Median :11.712   Median :19.02
##  Mean   :17.13   Mean   :17.51   Mean   :11.717   Mean   :19.02
##  3rd Qu.:17.80   3rd Qu.:18.18   3rd Qu.:12.391   3rd Qu.:19.71
##  Max.   :21.43   Max.   :21.54   Max.   :15.419   Max.   :23.81
```

```
##  NAs    :676     NAs    :796     NAs    :937      NAs    :1048
##  measurement_9    measurement_10   measurement_11  measurement_12
##  Min.   : 7.537   Min.   : 9.323   Min.   :12.46   Min.   : 5.167
##  1st Qu.:10.757   1st Qu.:15.209   1st Qu.:18.17   1st Qu.:10.703
##  Median :11.430   Median :16.127   Median :19.21   Median :11.717
##  Mean   :11.431   Mean   :16.118   Mean   :19.17   Mean   :11.702
##  3rd Qu.:12.102   3rd Qu.:17.025   3rd Qu.:20.21   3rd Qu.:12.709
##  Max.   :15.412   Max.   :22.479   Max.   :25.64   Max.   :17.663
##  NAs    :1227     NAs    :1300     NAs    :1468    NAs    :1601
##  measurement_13  measurement_14  measurement_15   measurement_16
##  Min.   :10.89   Min.   : 9.14   Min.   : 9.104   Min.   : 9.701
##  1st Qu.:14.89   1st Qu.:15.06   1st Qu.:13.957   1st Qu.:15.268
##  Median :15.63   Median :16.04   Median :14.969   Median :16.436
##  Mean   :15.65   Mean   :16.05   Mean   :14.996   Mean   :16.461
##  3rd Qu.:16.37   3rd Qu.:17.08   3rd Qu.:16.018   3rd Qu.:17.628
##  Max.   :22.71   Max.   :22.30   Max.   :21.626   Max.   :24.094
##  NAs    :1774    NAs    :1874    NAs    :2009     NAs    :2110
##  measurement_17        failure
##  Min.   : 196.8   Length   :26570
##  1st Qu.: 619.0   N.unique :    2
##  Median : 701.0   N.blank  :    0
##  Mean   : 701.3   Min.nchar:    2
##  3rd Qu.: 784.1   Max.nchar:    3
##  Max.   :1312.8
##  NAs    :2284
```

Delete ID column

```
train = train %>% select(- id)
```

Convert character variables to factors

```
train = train %>% mutate_if(is.character, as_factor)
```

Convert other categorical variables to factors

```
train <- train %>%
  mutate(
    attribute_2 = as.factor(attribute_2),
    attribute_3 = as.factor(attribute_3)
  )
```

Clean data, remove outliers

```
train = train %>% filter(measurement_2 < 22)
train = train %>% filter(measurement_8 < 23.5)
train = train %>% filter(measurement_14 > 8)
train = train %>% filter(measurement_15 < 23)
train = train %>% filter(measurement_17 < 1500)
```

Select variables of interest

```
train = train %>% select(c("loading", "measurement_17", "measurement_2", "mea
surement_5", "measurement_6", "measurement_8", "failure"))
```

Check for missing data

```
#skim(train)
```

Imputation

```
set.seed(1234)
imp_train = mice(train, m=5, method="pmm", printFlag=FALSE)

train_complete = complete(imp_train)
summary(train_complete)
```

```
##     loading       measurement_17   measurement_2    measurement_5
##  Min.   : 40.81   Min.   : 196.8   Min.   : 0.000   Min.   :12.07
##  1st Qu.:100.13   1st Qu.: 618.6   1st Qu.: 4.000   1st Qu.:16.45
##  Median :122.46   Median : 700.8   Median : 6.000   Median :17.13
##  Mean   :127.84   Mean   : 701.3   Mean   : 6.251   Mean   :17.13
##  3rd Qu.:149.07   3rd Qu.: 783.4   3rd Qu.: 8.000   3rd Qu.:17.81
##  Max.   :385.86   Max.   :1312.8   Max.   :21.000   Max.   :21.43
##  measurement_6   measurement_8   failure
##  Min.   :12.71   Min.   :15.27   No :15803
##  1st Qu.:16.83   1st Qu.:18.34   Yes: 4240
##  Median :17.52   Median :19.02
##  Mean   :17.51   Mean   :19.02
##  3rd Qu.:18.17   3rd Qu.:19.71
##  Max.   :21.54   Max.   :23.33
```

Split data

```
set.seed(123)
train_split = initial_split(train_complete, prop=0.7, strata=failure)
train_set = training(train_split)
test_set = testing(train_split)
```

Set up folds for cross-validation

```
set.seed(123)
rf_folds = vfold_cv(train_set, v = 5)
```

Random forest, R-defined tuning grid

```
train_recipe = recipe(failure ~ loading, train_set) %>%
  step_dummy(all_nominal(), -all_outcomes())

rf_model = rand_forest(mtry = tune(), min_n = tune(), trees = 100) %>%
  set_engine("ranger", importance = "permutation") %>%
  set_mode("classification")
```

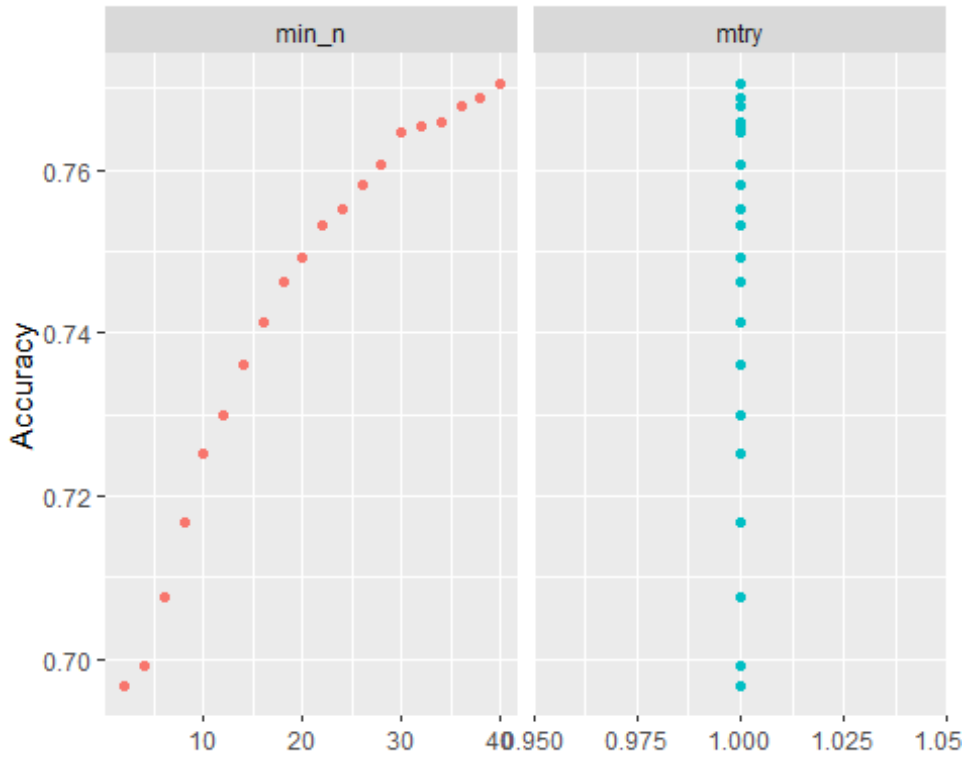
```
train_wflow =
  workflow() %>%
  add_model(rf_model) %>%
  add_recipe(train_recipe)

set.seed(123)
rf_res = tune_grid(
  train_wflow,
  resamples = rf_folds,
  grid = 20
)

## i Creating pre-processing data to finalize 1 unknown parameter: "mtry"
```

Evaluate parameter performance

```
rf_res %>%
  collect_metrics() %>%
  filter(.metric == "accuracy") %>%
  select(mean, min_n, mtry) %>%
  pivot_longer(min_n:mtry,
    values_to = "value",
    names_to = "parameter"
  ) %>%
  ggplot(aes(value, mean, color = parameter)) +
  geom_point(show.legend = FALSE) +
  facet_wrap(~parameter, scales = "free_x") +
  labs(x = NULL, y = "Accuracy")
```



Refine parameters with manual tuning

```
train_recipe = recipe(failure ~ loading, train_set) %>%
  step_dummy(all_nominal(), -all_outcomes())

rf_model = rand_forest(mtry = tune(), min_n = tune(), trees = 100) %>%
  set_engine("ranger", importance = "permutation") %>%
  set_mode("classification")

train_wflow =
  workflow() %>%
  add_model(rf_model) %>%
  add_recipe(train_recipe)

rf_grid = grid_regular(
  mtry(range = c(1,6)),
  min_n(range = c(2,40)),
  levels = 5
)

set.seed(123)
rf_res_tuned = tune_grid(
  train_wflow,
  resamples = rf_folds,
  grid = rf_grid
)
```

```
## → A | warning: ! 2 columns were requested but there was 1 predictor in the
 data.
##                ℹ 1 predictor will be used.

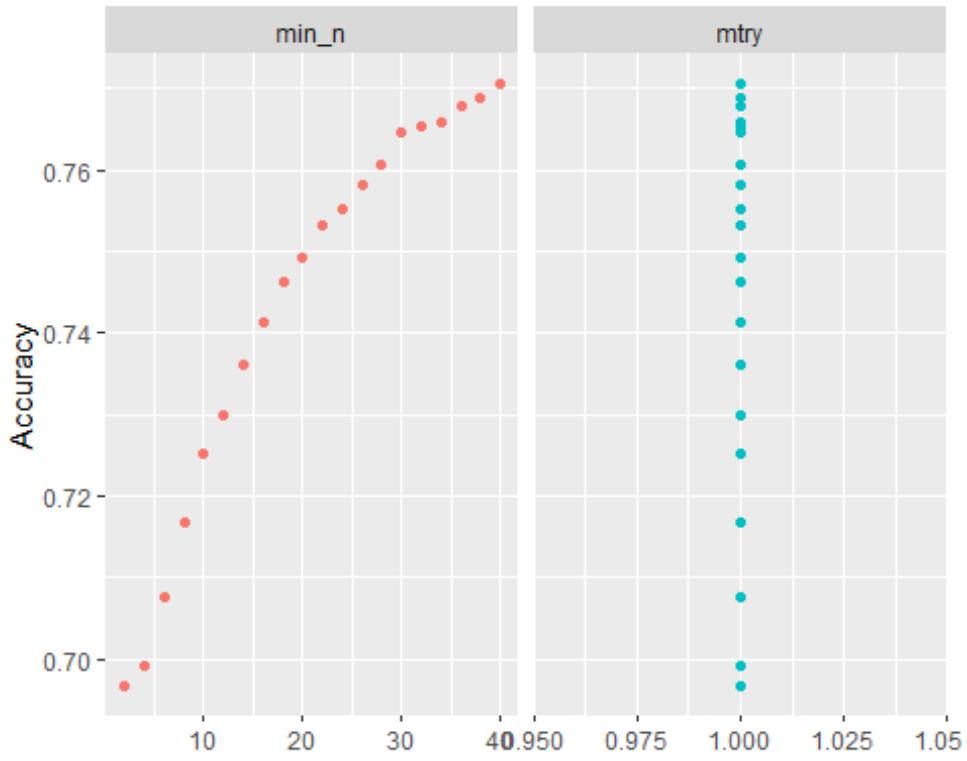
## There were issues with some computations   A: x1
                       → B | warning: ! 3 columns were requested but there wa
s 1 predictor in the data.
##                ℹ 1 predictor will be used.
## There were issues with some computations   A: x1There were issues with som
e computations   A: x2   B: x1
          → C | warning: ! 4 columns were requested but there was 1 predictor
 in the data.
##                ℹ 1 predictor will be used.
## There were issues with some computations   A: x2   B: x1There were issues
with some computations   A: x3   B: x2   C: x1
                                    → D | warning: ! 6 columns were requested b
ut there was 1 predictor in the data.
##                ℹ 1 predictor will be used.
## There were issues with some computations   A: x3   B: x2   C: x1There were
 issues with some computations   A: x4   B: x3   C: x2   D: x1There were issu
es with some computations   A: x6   B: x4   C: x3   D: x2There were issues wi
th some computations   A: x8   B: x6   C: x4   D: x3There were issues with so
me computations   A: x10   B: x8   C: x6   D: x4There were issues with some c
omputations   A: x12   B: x10   C: x8   D: x6There were issues with some comp
utations   A: x15   B: x12   C: x10   D: x8There were issues with some comput
ations   A: x18   B: x15   C: x12   D: x10There were issues with some computa
tions   A: x21   B: x18   C: x15   D: x12There were issues with some computat
ions   A: x24   B: x21   C: x18   D: x15There were issues with some computati
ons   A: x28   B: x24   C: x21   D: x18There were issues with some computatio
ns   A: x32   B: x28   C: x24   D: x21There were issues with some computation
s   A: x36   B: x32   C: x28   D: x24There were issues with some computations
   A: x40   B: x36   C: x32   D: x28There were issues with some computations
  A: x45   B: x40   C: x36   D: x32There were issues with some computations
 A: x50   B: x45   C: x40   D: x36There were issues with some computations
A: x55   B: x50   C: x45   D: x40There were issues with some computations
A: x60   B: x55   C: x50   D: x45There were issues with some computations
A: x61   B: x55   C: x50   D: x45There were issues with some computations
A: x62   B: x56   C: x50   D: x45There were issues with some computations
A: x63   B: x57   C: x51   D: x45There were issues with some computations
A: x64   B: x58   C: x52   D: x46There were issues with some computations
A: x66   B: x59   C: x53   D: x47There were issues with some computations
A: x68   B: x61   C: x54   D: x48There were issues with some computations
A: x70   B: x63   C: x56   D: x49There were issues with some computations
A: x72   B: x65   C: x58   D: x51There were issues with some computations
A: x75   B: x67   C: x60   D: x53There were issues with some computations
A: x78   B: x70   C: x62   D: x55There were issues with some computations
A: x81   B: x73   C: x65   D: x57There were issues with some computations
A: x84   B: x76   C: x68   D: x60There were issues with some computations
A: x88   B: x79   C: x71   D: x63There were issues with some computations
A: x92   B: x83   C: x74   D: x66There were issues with some computations
```

A: x96 B: x87 C: x78 D: x69There were issues with some computations A: x100 B: x91 C: x82 D: x73There were issues with some computations A: x105 B: x95 C: x86 D: x77There were issues with some computations A: x110 B: x100 C: x90 D: x81There were issues with some computations A: x115 B: x105 C: x95 D: x85There were issues with some computations A: x120 B: x110 C: x100 D: x90There were issues with some computations A: x121 B: x110 C: x100 D: x90There were issues with some computations A: x122 B: x111 C: x100 D: x90There were issues with some computations A: x123 B: x112 C: x101 D: x90There were issues with some computations A: x124 B: x113 C: x102 D: x91There were issues with some computations A: x126 B: x114 C: x103 D: x92There were issues with some computations A: x128 B: x116 C: x104 D: x93There were issues with some computations A: x130 B: x118 C: x106 D: x94There were issues with some computations A: x132 B: x120 C: x108 D: x96There were issues with some computations A: x135 B: x122 C: x110 D: x98There were issues with some computations A: x138 B: x125 C: x112 D: x100There were issues with some computations A: x141 B: x128 C: x115 D: x102There were issues with some computations A: x144 B: x131 C: x118 D: x105There were issues with some computations A: x148 B: x134 C: x121 D: x108There were issues with some computations A: x152 B: x138 C: x124 D: x111There were issues with some computations A: x156 B: x142 C: x128 D: x114There were issues with some computations A: x160 B: x146 C: x132 D: x118There were issues with some computations A: x165 B: x150 C: x136 D: x122There were issues with some computations A: x170 B: x155 C: x140 D: x126There were issues with some computations A: x175 B: x160 C: x145 D: x130There were issues with some computations A: x180 B: x165 C: x150 D: x135There were issues with some computations A: x181 B: x165 C: x150 D: x135There were issues with some computations A: x182 B: x166 C: x150 D: x135There were issues with some computations A: x183 B: x167 C: x151 D: x135There were issues with some computations A: x184 B: x168 C: x152 D: x136There were issues with some computations A: x186 B: x169 C: x153 D: x137There were issues with some computations A: x188 B: x171 C: x154 D: x138There were issues with some computations A: x190 B: x173 C: x156 D: x139There were issues with some computations A: x192 B: x175 C: x158 D: x141There were issues with some computations A: x195 B: x177 C: x160 D: x143There were issues with some computations A: x198 B: x180 C: x162 D: x145There were issues with some computations A: x201 B: x183 C: x165 D: x147There were issues with some computations A: x204 B: x186 C: x168 D: x150There were issues with some computations A: x208 B: x189 C: x171 D: x153There were issues with some computations A: x212 B: x193 C: x174 D: x156There were issues with some computations A: x216 B: x197 C: x178 D: x159There were issues with some computations A: x220 B: x201 C: x182 D: x163There were issues with some computations A: x225 B: x205 C: x186 D: x167There were issues with some computations A: x230 B: x210 C: x190 D: x171There were issues with some computations A: x235 B: x215 C: x195 D: x175There were issues with some computations A: x240 B: x220 C: x200 D: x180There were issues with some computations A: x241 B: x220 C: x200 D: x180There were issues with some computations A: x242 B: x221 C: x200 D: x180There were issues with some computations A: x243 B: x222 C: x201

```
 D: x180There were issues with some computations   A: x244   B: x223   C: x20
2   D: x181There were issues with some computations   A: x246   B: x224   C:
x203   D: x182There were issues with some computations   A: x248   B: x226
C: x204   D: x183There were issues with some computations   A: x250   B: x228
   C: x206   D: x184There were issues with some computations   A: x252   B: x
230   C: x208   D: x186There were issues with some computations   A: x255
B: x232   C: x210   D: x188There were issues with some computations   A: x258
   B: x235   C: x212   D: x190There were issues with some computations   A: x
261   B: x238   C: x215   D: x192There were issues with some computations
A: x264   B: x241   C: x218   D: x195There were issues with some computations
   A: x268   B: x244   C: x221   D: x198There were issues with some computati
ons   A: x272   B: x248   C: x224   D: x201There were issues with some comput
ations   A: x276   B: x252   C: x228   D: x204There were issues with some com
putations   A: x280   B: x256   C: x232   D: x208There were issues with some
computations   A: x285   B: x260   C: x236   D: x212There were issues with so
me computations   A: x290   B: x265   C: x240   D: x216There were issues with
 some computations   A: x295   B: x270   C: x245   D: x220There were issues w
ith some computations   A: x300   B: x275   C: x250   D: x225There were issue
s with some computations   A: x300   B: x275   C: x250   D: x225
```

Evaluate parameter performance

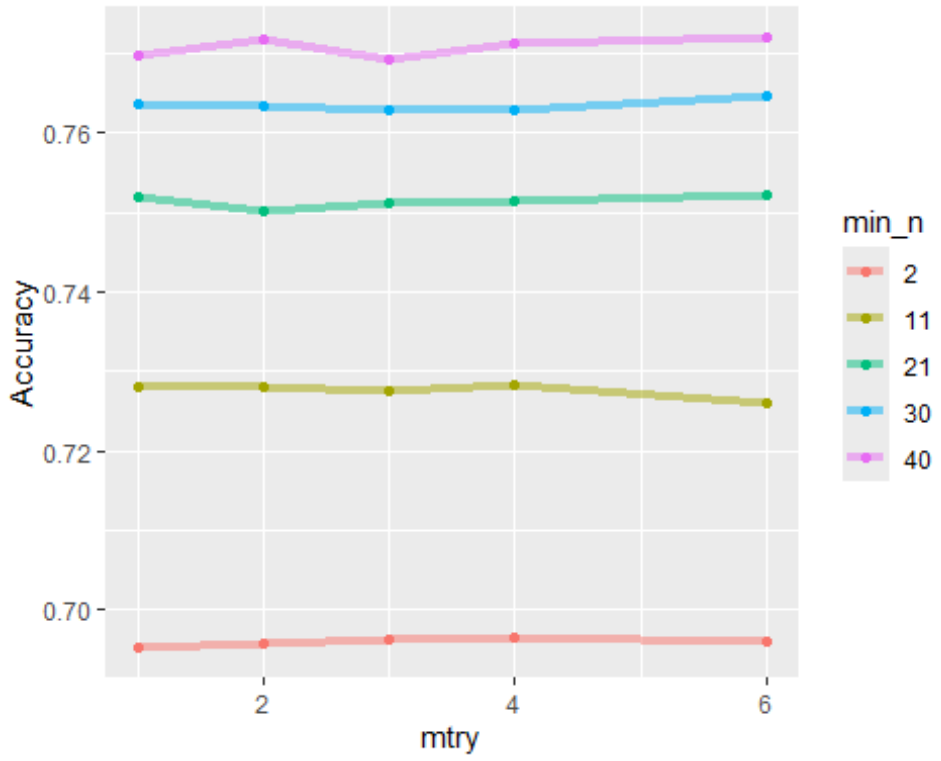
```
rf_res %>%
  collect_metrics() %>%
  filter(.metric == "accuracy") %>%
  select(mean, min_n, mtry) %>%
  pivot_longer(min_n:mtry,
    values_to = "value",
    names_to = "parameter"
  ) %>%
  ggplot(aes(value, mean, color = parameter)) +
  geom_point(show.legend = FALSE) +
  facet_wrap(~parameter, scales = "free_x") +
  labs(x = NULL, y = "Accuracy")
```



Alternate parameter view

```
rf_res_tuned %>%
  collect_metrics() %>%
  filter(.metric == "accuracy") %>%
  mutate(min_n = factor(min_n)) %>%
  ggplot(aes(mtry, mean, color = min_n)) +
  geom_line(alpha = 0.5, size = 1.5) +
  geom_point() +
  labs(y = "Accuracy")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## ℹ Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Extract best parameters

```
best_params <- select_best(rf_res_tuned, metric = "accuracy")
```

Finalize model

```
final_rf_model <- finalize_model(rf_model, best_params)
```

Finalize workflow

```
final_wflow <- finalize_workflow(train_wflow, best_params)
```

Fit to train_set

```
final_fit <- fit(final_wflow, data = train_set)
```

```
## Warning: ! 6 columns were requested but there was 1 predictor in the data.
## ℹ 1 predictor will be used.
```

Predictions for train_set

```
trainpredrf = predict(final_fit, train_set)
head(trainpredrf)
```

```
## # A tibble: 6 × 1
##   .pred_class
##   <fct>
## 1 No
## 2 No
```

```
## 3 No
## 4 No
## 5 No
## 6 No
```

Confusion matrix for train_set

```
confusionMatrix(trainpredrf$.pred_class, train_set$failure, positive = "Yes")
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    No   Yes
##        No  10983  2543
##        Yes    79   425
##
##                Accuracy : 0.8131
##                  95% CI : (0.8066, 0.8195)
##     No Information Rate : 0.7885
##     P-Value [Acc > NIR] : 2.121e-13
##
##                   Kappa : 0.1954
##
##  Mcnemar's Test P-Value : < 2.2e-16
##
##             Sensitivity : 0.14319
##             Specificity : 0.99286
##          Pos Pred Value : 0.84325
##          Neg Pred Value : 0.81199
##              Prevalence : 0.21155
##          Detection Rate : 0.03029
##    Detection Prevalence : 0.03592
##       Balanced Accuracy : 0.56803
##
##        'Positive' Class : Yes
##
```

Predictions on test set

```
testpredrf = predict(final_fit, test_set)
head(testpredrf)
```

```
## # A tibble: 6 × 1
##   .pred_class
##   <fct>
## 1 No
## 2 No
## 3 No
## 4 No
## 5 No
## 6 No
```

```r
confusionMatrix(testpredrf$.pred_class, test_set$failure, positive = "Yes")
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction   No  Yes
##        No  4548 1209
##        Yes  193   63
##
##                Accuracy : 0.7668
##                  95% CI : (0.7559, 0.7775)
##     No Information Rate : 0.7885
##     P-Value [Acc > NIR] : 1
##
##                   Kappa : 0.0125
##
##  Mcnemar's Test P-Value : <2e-16
##
##             Sensitivity : 0.04953
##             Specificity : 0.95929
##          Pos Pred Value : 0.24609
##          Neg Pred Value : 0.78999
##              Prevalence : 0.21154
##          Detection Rate : 0.01048
##    Detection Prevalence : 0.04257
##       Balanced Accuracy : 0.50441
##
##        'Positive' Class : Yes
##
```

Save the file

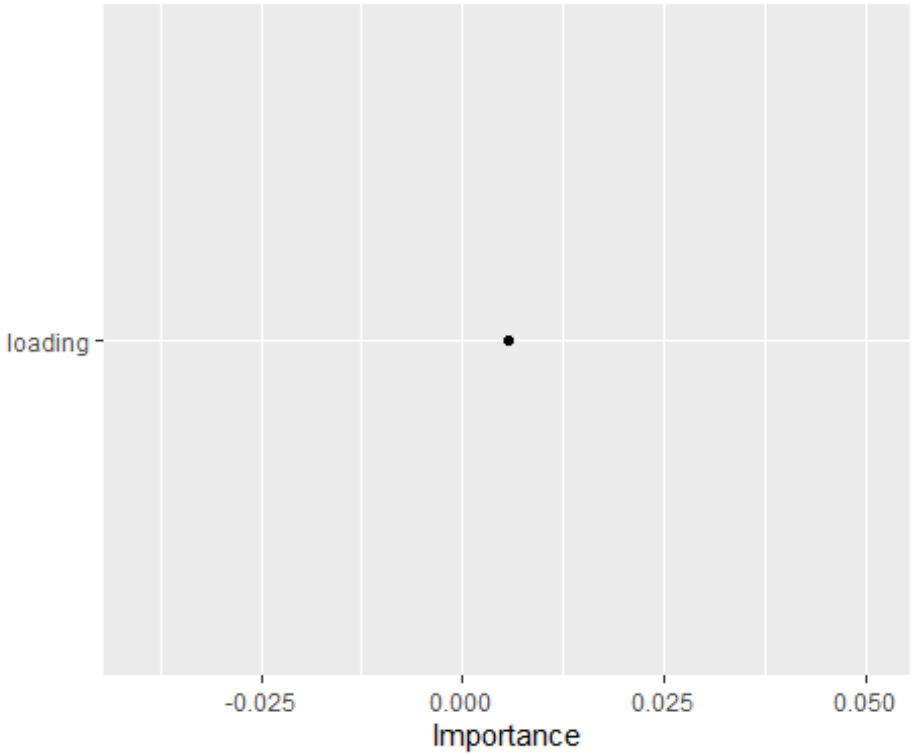
```r
saveRDS(final_fit, "final_fit.rds")
```

Load the model

```r
final_fit = readRDS("final_fit.rds")
```

Variable importance

```r
final_fit %>% extract_fit_parsnip() %>% vip(geom = "point")
```


loading
-0.025
0.000
0.025
0.050
Importance